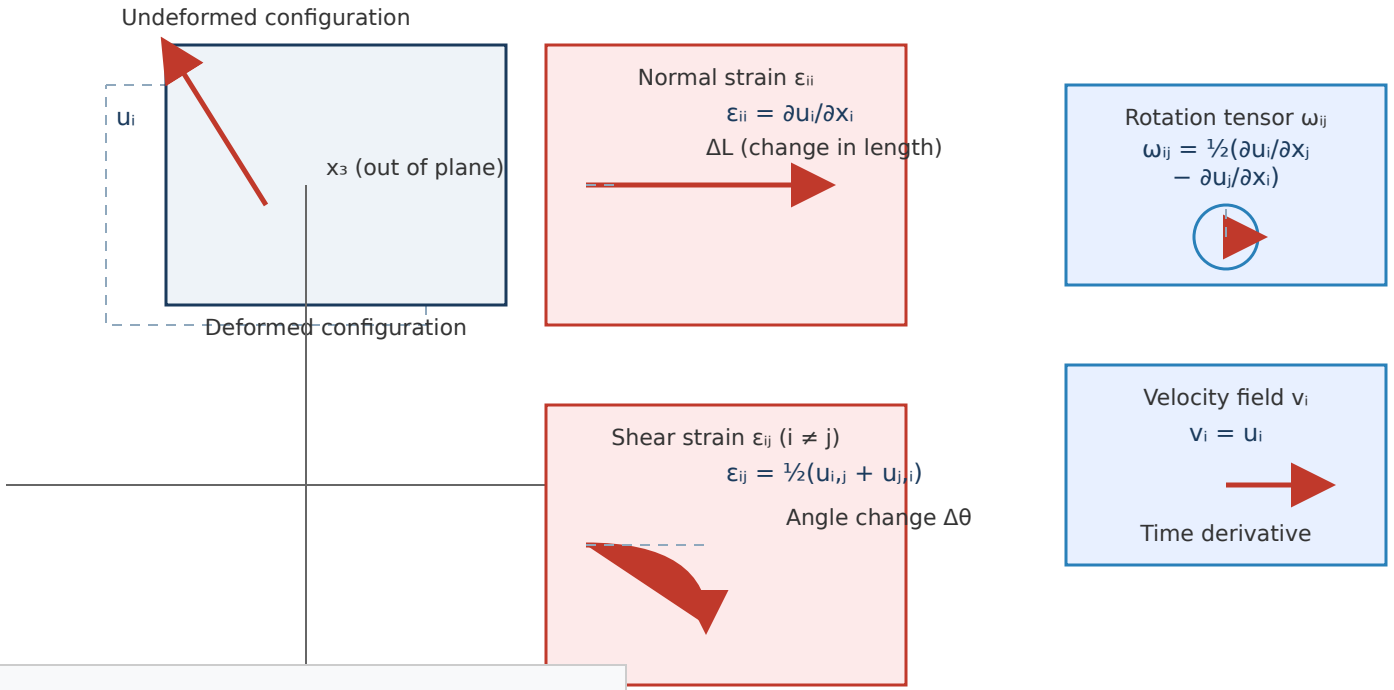
Fundamental continuum kinematics

Displacement, strain, rotation, and velocity in an infinitesimal continuum element



Key definitions:

$u_{i,j} = \varepsilon_{ij} + \omega_{ij}$ (displacement gradient decomposition)

$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$ (infinitesimal strain)

$\omega_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i})$ (rotation tensor)

$v_i = u_i$ (velocity)

Infinitesimal strain assumption: $\|\varepsilon_{ij}\| \ll 1$. Valid for linearized wave propagation.